MIL-A-23282B(Wep)
19 April 1965
Superseding
MIL-A-23282A (Wep)
23 October 1962

MILITARY SPECIFICATION

ACTUATOR

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification, together with BUWEPS LD 495974, establishes the requirements for the procurement of a magnetic particle clutch actuator which supplies course control to Torpedo Mk 37 Mods 0 and 1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-325	Zinc Plating (Electrodeposited)
QQ-S-571	Solder, Lead Alloy, Tin Lead Alloy and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-P-6808	Primer Coating, Zinc Chromate, for Aircraft use, Application of
MIL-S-6872	Soldering Process, General Specifications for

FSC 1355

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MIL-P-8585	Primer, Zinc Chromate, Low-moisture Sensitivity
MIL-A-8625	Anodized Coatings, for Aluminum and Aluminum Alloys
MIL-C-13335	Coatings for Magnesium and Magnesium Alloys
MIL-P-16298	Electric Machines Having Rotating Parts and Associated Repair Parts; Packing of
MIL-W-16878	Wire, Electrical, Insulated, High Temperature

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 495974	Actuator, Course Control
BUWEPS LD 495980	Shipping (Packing) Condition (for one Actuator LD 495974; 694739-1)
BUWEPS LD 620137	Shipping (Packing) Condition (for one Actuator LD 495974; 694739-2)
BUWEPS Dwg 694739	Actuator, Course Control
BUWEPS Dwg 877336	Actuator, Adjustment Set Assembled
BUWEPS Dwg 877495	Servo Actuator Adjustment and Test Stand Assembled

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PUBLICATIONS

NAVY DEPARTMENT

NAVORD OD 12301	Insulation Resistance and High Potential Measurements and Procedures
WR-11	Design and Test of Packaging, Packing, Shipping, and Handling Equipment for Weapon System Components

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C.)

3. REQUIREMENTS

3.1 Description. The magnetic particle clutch actuator covered by this specification operates the torpedo control surface through an appropriate linkage. Input power is supplied from the torpedo drive shaft by means of toothed pulleys and belt to two magnetic particle clutches which derive their control power from the servoamplifier. The clutches drive a gear reduction unit and in turn an output arm through which the control surface positions are accomplished. Incorporated in the gear reduction unit is a dual potentiometer, one section of which supplies an actuator

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position signal to the servoamplifier, and the other section, a control surface position signal to the recording equipment. A tachometer is geared to the clutch output shaft, and applies a position rate signal to the servoamplifier.

3.2 General requirements.

3.2.1 Contractor-furnished data. Data required by this specification, or by applicable documents referenced in Section 2, need not be furnished unless specified in the contract or order (see 6.2.4).

3.2.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the actuator shall be manufactured using the methods, materials, and procedures proposed for the production. The sample will be tested as specified in Section 4 herein, and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the actuator shall conform strictly to the detailed requirements of this specification and applicable drawings.

3.2.3.1 Corrosion resistance. Material shall be of a corrosion resisting type or shall be suitably processed to resist corrosion.

3.2.3.2 Standard parts and commercial items. The greatest practical use shall be made of standard parts that are commensurate with design requirements. Such parts shall be selected from Military Standards and Military Specifications, in that order of preference.

3.2.3.3 Hookup wire. All hookup wire used in the actuator shall conform to Specification MIL-W-16878, unless otherwise specified.

3.2.3.4 Ferrous metal parts. Electrical sheet steel assemblies shall be processed by accepted commercial methods to provide protection against corrosion. Where other non-corrosive-resistant ferrous metals must be used, the exposed surfaces shall be zinc plated in accordance with Specification QQ-Z-325, Type II, Class 3 (0.0002 minimum thickness).

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3.2.3.5 Aluminum alloy parts. Aluminum and aluminum alloys, if used, shall be anodized in accordance with Specification MIL-A-8625, bichromate seal, or chemically treated in accordance with Specification MIL-C-5541.

3.2.3.6 Magnesium alloy parts. Magnesium and magnesium alloys, if used, shall be treated in accordance with Specification MIL-C-13335 and given one coat of Primer, Specification MIL-P-8585, in accordance with Specification MIL-P-6808.

3.2.3.7 Soldering. Only class SN60 solder, in accordance with Specification QQ-S-571, shall be used for electrical connections. Soldering processes shall not depend solely on soft solder for mechanical strength except for relatively light parts which have proved suitable in actual use, Soldering processes shall conform to requirements of Specification MIL-S-6872.

3.2.3.8 Actuator weight. The weight of the actuator shall not exceed seven pounds.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the actuator shall be in accordance with BUWEPS Drawing 694739.

3.2.5 Moisture, fungus, and corrosion. The actuator shall be rendered moisture, fungus, and corrosion resistant as required by BUWEPS LD 495974 and related drawings and specifications.

3.2.6 Marking. Marking shall be in accordance with BUWEPS Drawing 694739.

3.2.7 Interchangeability. Individual parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with mating parts of the unit and other like units. If deviations greater or smaller than the tolerances shown on the drawings or specifications are required for interchangeability, or if hand fitting beyond that shown on the applicable drawings is required, then changes in dimensions are necessary and revision and correction of the applicable drawings or specifications is required. See 3.2.10.

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3.2.8 Gages. The contractor shall provide himself with whatever gages are necessary to ensure that the material to which this specification applies will meet the dimension requirements shown on the applicable drawing. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.9 Threads. Unless otherwise indicated, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.10 Conflicting requirements . Conflicting requirements arising between this specification and any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Performance requirements and product characteristics. The magnetic particle clutch actuator shall meet the following performance requirements and product characteristics.

3.3.1 Insulation resistance. The insulation resistance, when measured between all pins of P40 (BUWEPS Drawing 694739) and an uninsulated metallic part of the actuator, shall be 20 megohms minimum at ambient temperature and 1 megohm minimum at the conclusion of a duty cycle, as specified in 3.3.10, with 500 plus or minus 20 volts direct current (de) impressed across the insulation.

3.3.2 Magnetic particle clutch coils

3.3.2.1 Clutch circuit resistance. Each magnetic particle clutch circuit shall have a dc resistance of 5720 ohms plus or minus 7-1/2 percent, corrected to 25 degrees centigrade (C) (77 degrees Fahrenheit (F)) when tested as in 4.5.4.2.

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3.3.2.2 Actuator circuit. Each actuator shall conform to the wiring diagram (BUWEPS Drawing 694739) when tested as in 4.5.4.2.

3.3.2.3 Diode. The following specifications apply to the diodes delineated on BUWEPS Drawing 694739 Sheet 2.

(a) Type: Silicon.

(b) Peak inverse voltage: 300 volts minimum at 100 degrees C.

(c) Average rectified current at 100 degrees C: 100 milliamps minimum.

(d) Surge current for 10 milliseconds: 2 amps minimum.

(e) Forward voltage drop, full cycle average at 100 degrees C and 100 milliamps: 1000 millivolts (rev) maximum.

(f) Reverse leakage current at 100 degrees C and 300 volts reverse, full cycle average: 25 microamps maximum.

(g) Operating temperature: minus 50 degrees C to plus 150 degrees C minimum.

(h) Storage temperature: minus 50 degrees C to plus 150 degrees C minimum.

(i) The diode selected shall be capable of meeting all other requirements of this- specification."

3.3.3 Binding.

3.3.3.1 Input. With the actuator in a horizontal position and with the clutch deenergized, there shall be no evidence of binding or tightness when the input shaft is manually rotated, when tested as in 4.5.4.3.1.

3.3.3.2 Output. With the actuator in a horizontal position and with the clutch deenergized, a maximum breakaway torque of 20 pound-inches shall be required to move the output arm to any position within the limits of travel specified in 3.3.4, when tested as in 4.5.4.3.2.

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3.3.4 Output travel. The total travel from stop to stop of the output arm (BUWEPS Drawing 694739) shall be 28 plus or minus 1 degree when tested as in 4.5.4.4.

3.3.4.1 Tachometer. A tachometer shall be included as an integral component of the output drive. The amplitude of the tachometer output shall be proportional to the rotational speed of the output shaft. The output polarity of the tachometer shall be in accordance with Figure 1 of this specification and the requirements of BUWEPS Drawing 694739 Sheet 2. The tachometer shall be located so that the output shall not be less than 6.7 volts or more than 8.1 volts for the maximum output position of the typical voltage output trace of the tachometer, Figure 1. This output voltage applies for an input pulley speed of 5700 rpm.

3.3.4.2 The following specifications apply to the tachometer delineated on BUWEPS Drawing 694739 Sheet 2.

(a) Permament magnet field.

(b) 2.2 to 2.8 volts per 1000 rpm with a 1000 to 1040 ohm load; 3.0 volts per 1000 rpm with no load (reference).

(c) Output applies to clockwise and counterclockwise rotation.

(d) Maximum load 0.1 watt. Average load 0.04 watt.

(e) 0.04 watt output with one reversal per second average.

(f) Ripple factor shall not exceed plus or minus 5 percent.

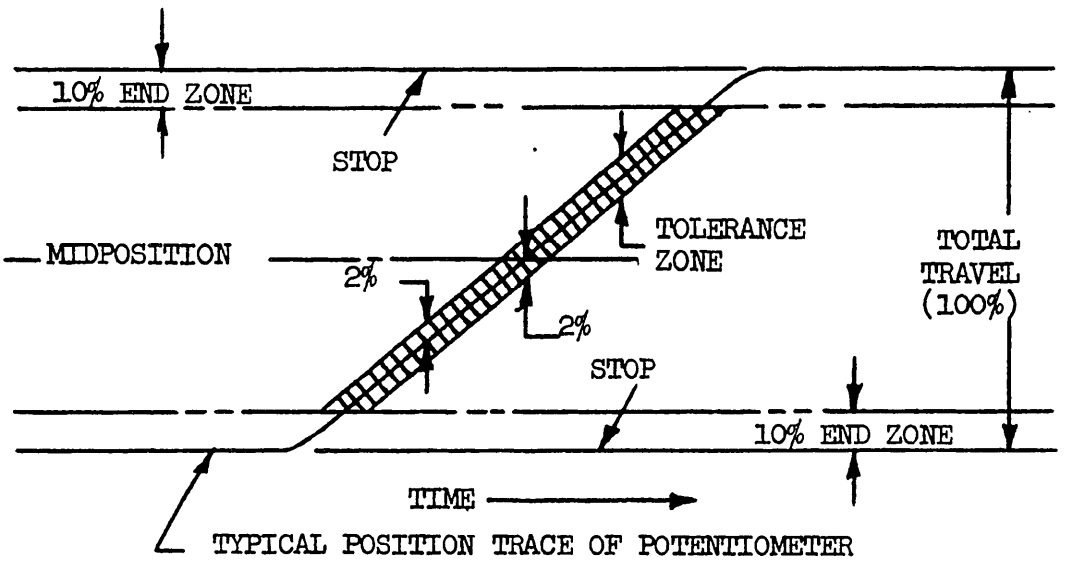
(g) The tachometer selected shall be capable of meeting all other requirements of this specification.

3.3.4.2.1 In a balanced current condition at input speed of 5700 plus or minus 5 percent rpm, the ripple component of the output voltage shall not exceed 80 millivolts peak-to-peak.

3.3.4.3 Output shaft. The output shaft. speed shall be proportional to the input pulley speed and shall have a speed reduction ratio of not less than 263 to 1 and not greater than 285 to 1.

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LINEARITY & SMOOTHNESS OF OPERATION



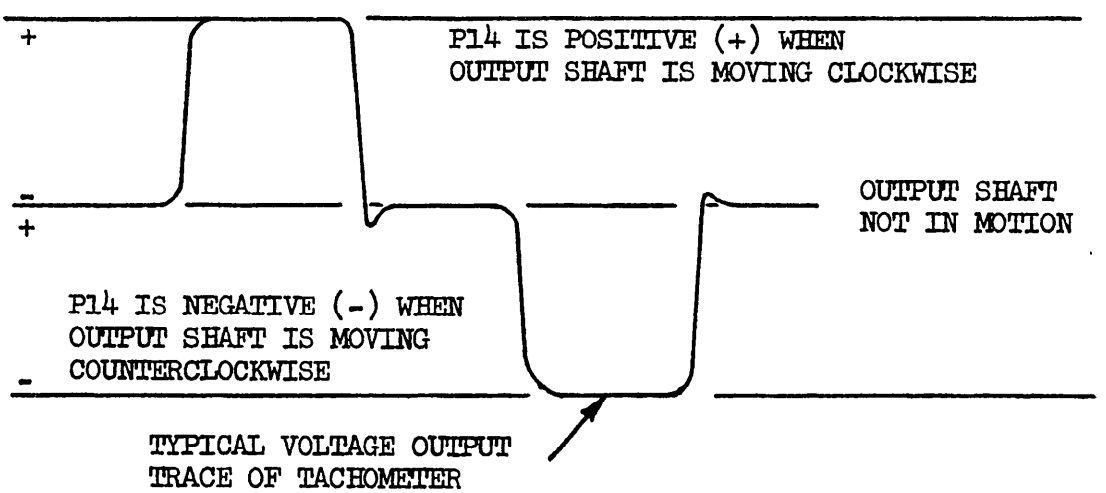


FIGURE 1

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3.3.5 Backlash

3.3.5.1 Output arm. The backlash of the output arm shall not exceed 0 degrees, 30 minutes, with the input pulley held stationary, when tested as in 4.5.4.5.

3.3.5.2 Potentiometer shaft. The backlash of the potentiometer shaft shall not exceed 0 degrees, 30 minutes.

3.3.6 Output direction

3.3.6.1 Clockwise. The actuator output arm shall move clockwise (cw) (viewed facing the output shaft) when the input pulley is driven cw (viewed facing the pulley) and a current of 7 plus or minus 2 milliamperes (ma) dc flows between terminals 1 and 2 of connector P40 (BUWEPS Drawing 694739), energizing the aft magnetic particle clutch coil. Test as in 4.5.4.6.1.

3.3.6.2 Counterclockwise. The actuator arm shall move counterclockwise (ccw) when the input pulley is driven cw and a current of 7 plus or minus 2 ma dc flows between terminals 2 and 3 of P40 (BUWEPS Drawing 694739), energizing the forward magnetic particle clutch coil. Test as in 4.5.4.6.2.

3.3.7 Input speed. The actuator shall be capable of meeting the requirements of 3.3.8 and 3.3.9 with input pulley speeds of 1000 to 6500 rpm, when tested as in 4.5.4.7, 4.5.4.8, 4.5.4.9, 4.5.4.10, and 4.5.4.11.

3.3.8 Output torque

3.3.8.1 The actuator shall produce an output torque of 190 plus or minus 60 pounds-inches when either magnetic particle clutch is energized with a control current of 12.5 plus or minus 0.5 ma.

3.3.8.2 Balanced current operation. When control current is applied to both coils so that the sum of the currents is 11 plus or minus 2.5 ma and the individual currents are equal within 0.25 ma, the average torque at the output arm, in either direction, shall not exceed 15 pounds-inches. Torque shall not exceed 20 pounds-inches at any time. The foregoing requirement is for the condition in which the “balanced current” described above is obtained by increasing the current

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in both clutches from zero condition, and the approach from zero condition to balance current is made without overshoot, i.e., without reversing the direction of the charge of the current.

3.3.9 Smoothness of operation

3.3.9.1 Unloaded output. The motion of the unloaded actuator output shall be linear, on a time base, within plus or minus 2 percent of total travel, excluding 10 percent of the angular travel at the start from either end, when the magnetic particle clutches are alternately energized with 7 plus or minus 2 ma. (See Figure 1.) Test as in 4.5.4.9.1.

3.3.9.2 Loaded output at rated current. The output arm shall not oscillate more than 2 degrees after the actuator has reached its loaded position (in either direction), when either magnetic particle clutch is energized by a constant current of 12.5 plus or minus 0.5 ma, and the other magnetic particle clutch has its current reduced to less than 1 ma, and when the load applied to the output arm has a spring rate of 28 plus or minus 1 pound-inch per degree. This requirement applies when the magnetic particle clutch coil energization is switched to this condition from any other immediately previous normal operating condition. Test as in 4.5.4.9.2.

3.3.9.3 Loaded output at balanced current. The output arm shall not oscillate more than 3 degrees when the control currents in each magnetic particle clutch are equalized and remain equalized within 0.25 ma, and each magnetic particle clutch has a control current of 5.75 plus or minus 0.75 ma, and when the load applied to the output arm has a spring rate of 28 plus or minus 1 pound-inch per degree. This requirement applies when the coil energization is switched to the above condition from any other immediately previous normal operation.

3.3.10 Duty cycle, defined. For the purpose of testing to simulate actuator service operation, a "duty cycle" is defined as a minimum of 30 minutes of continuous operation under the following conditions:

3.3.10.1 The input pulley shall be driven at 5000 to 6000 rpm in a clockwise direction when facing the pulley.

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3.3.10.2 The actuator output arm shall drive a torque load having a spring rate of 28 plus or minus 1 pound-inch per degree.

3.3.10.3 The balance of current between the magnetic particle clutch shall vary according to the timing graph of one element of the duty cycle, as shown in Figure 2. During the time period labeled "balanced current", the control current in each magnetic particle clutch is at 5.75 ma plus or minus 0.75 ma, and the two magnetic particle clutch coil currents are equal within 0.25 ma.

3.3.10.4 Before the start of any duty cycle, the actuator temperature shall be close to ambient, as evidenced by one of the following conditions:

(a) The temperature of the hottest spot does not exceed ambient by more than 5 degrees C (9 degrees F).

(b) The actuator has been cooling continuously in ambient air for 50 minutes.

3.4 Environmental requirements

3.4.1 Vibration, transportation.

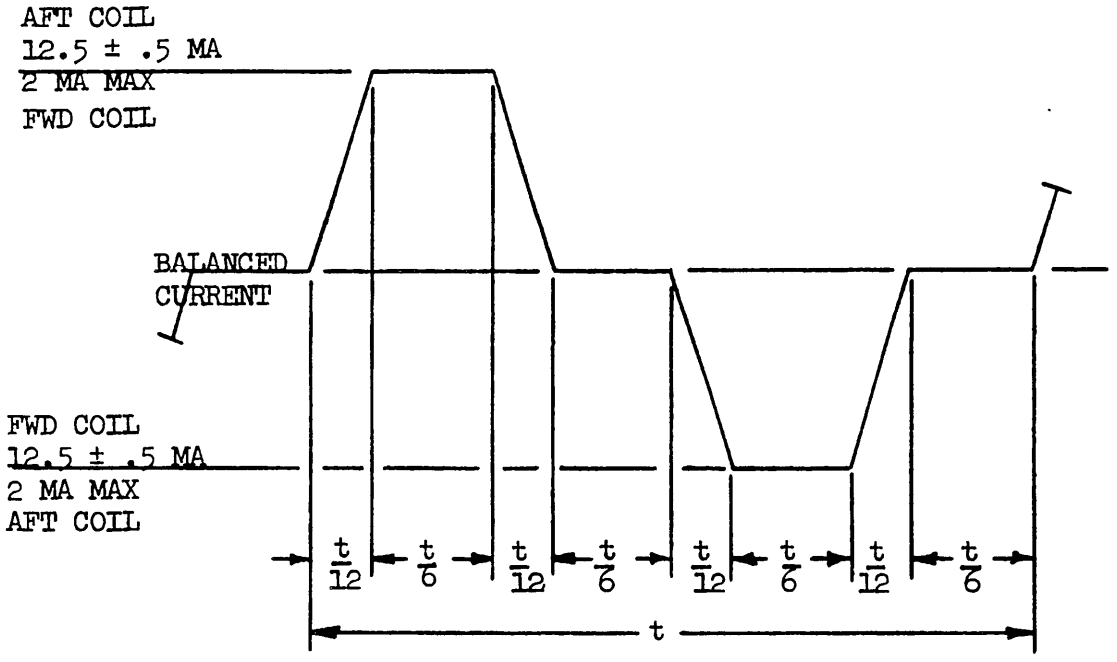
3.4.1.1 Level A (multiple pack). The actuator, packaged and packed in accordance with Section 5, Level A, shall be subjected to the vibration tests of 4.4.1.1. Thereafter, the packed condition shall be examined for structural integrity to assure safe delivery of the actuator.

3.4.1.2 Level C. The actuator, packaged and packed in accordance with Section 5, Level C, or the contract or the purchase order, shall be subjected to the vibration test of 4.4.1.2. Thereafter, the actuator shall show no evidence of damage and shall meet all the performance requirements of 3.3.

3.4.2 Vibration; ready, nonoperating. The actuator, less pack and packaging and mounted as specified in 4.4.2, shall be capable of withstanding vertical vibrations for a-period of 3 hours at a frequency of 60 cycles per second (cps) and a double amplitude (peak-to-peak) of 0.016 plus or minus 0.002 inch, and thereafter shall show no damage and shall meet the performance requirements and product characteristics of this specification.

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DUTY CYCLE



ONE ELEMENT OF A DUTY CYCLE WHERE: t MAY VARY FROM 6 TO 150 SECONDS AND WHERE THE DUTY CYCLE γ IS 1800 SECONDS MINIMUM.

ANY INCREMENTAL TIME MAY VARY BY 25% OF ITS OWN VALUE.

THE NUMBER OF ELEMENTS, N, TO COMPLETE A DUTY CYCLE: $N = \frac{\gamma}{t} = \frac{1800(MIN)}{t}$

FIGURE 2

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3.4.3 Shock; transportation.

3.4.3.1 Level A (multiple pack). The actuator, packaged and packed in accordance with Section 5, Level A, shall be subjected to the shock tests specified in 4.4.3.1. Thereafter, the pack shall be visually examined for structural integrity to assure safe delivery of the actuator.

3.4.3.2 Level C. The actuator, packaged and packed in accordance with Section 5, Level C, shall be subjected to the shock tests specified in 4.4.3.2. Thereafter, the actuator shall show no evidence of damage and shall meet all the performance requirements of 3.3.

3.4.4 Temperature and humidity

3.4.4.1 High temperature and humidity. The actuator, less pack and packaging, shall be capable of withstanding a temperature of plus 130 degrees F plus or minus 5 degrees F, 90 to 95 percent relative humidity, for a period of 48 hours in accordance with the test procedures specified in 4.4.4.1, and thereafter shall show no corrosion or damage and shall meet the performance requirements and product characteristics of this specification.

3.4.4.2 Low temperature. The actuator, less pack and packaging, shall be capable of withstanding a temperature of minus 20 degrees plus or minus 5 degrees F for a period of four hours in accordance with the test procedures specified in 4.4.4.2, and thereafter shall show no corrosion or damage and shall meet the performance requirements and product characteristics of this specification.

3.4,5 Operating life, The operating life of the actuator shall consist of a minimum of 50 duty cycles, as defined in 3.3.10. Any actuator under test must be cooled between duty cycles, as specified in 3.3.10.4.

3.4.5.1 At no time during the operating life tests shall the actuator torque exceed the limits specified in 3.3.8.1 or the oscillations exceed the limits of 3.3.9.2.

3.4.6 Storage life. When preserved and packaged as specified, the actuator shall meet all the performance requirements and product characteristics of this specification after being stored for a period of five years under temperature and humidity conditions as follows:

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3.4.6.1 Temperature. Minus 65 degrees F to plus 160 degrees F (minus 54 degrees C to plus 71 degrees C).

3.4.6.2 Humidity. Up to 95 percent relative (maximum).

3.5 Cleanup. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The actuator, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where-such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the magnetic particle clutch actuator shall be classified as follows:

a. Preproduction inspection (paragraph 4.4)
b. Periodic production inspection (paragraph 4.4)
c. Acceptance inspection (paragraph 4.5)

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4.2.1 Tests after delivery. Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein, as deemed necessary, after installation of the actuator in Torpedo Mk 37 Mods 0 and 1. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of five actuators manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Five diodes (3.3.2.3) and five tachometers (3.3.4.2), suitably packaged, shall also be submitted for determination of their compliance with their specific requirements. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the actuator by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. From each succeeding lot or grand lot of 300 or less, a random sample of five actuators, five tachometers, and five diodes, preserved, packaged, and packed as specified by the contract or order, will be selected by the Government Inspector for submission to a Bureau of Naval Weapons designated activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The production lot from which the periodic production sample was taken shall be rejected if any of the actuators, tachometers, or diodes of the sample lot fail to meet the specifications and pertinent drawing requirements, and fail the tests specified in 4.4 and 4.5. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.3.3 Acceptance sampling. The tests and examinations detailed in 4.5 shall be conducted on each actuator to be delivered; therefore, there shall be no sampling for acceptance. Defective units shall be rejected.

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4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the acceptance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Bureau of Naval Weapons.

4.4.1 Vibration; transportation

4.4.1.1 Level A (multiple pack). The actuator, packaged and packed in accordance with Section 5, Level A, shall be vibrated in accordance with WR-11. Starting at 2 cps gradually increase the frequency until the pack just begins to leave the table. At this point, vibrate the actuator for a period of one hour. Thereafter, the packed condition shall be examined for structural integrity to assure safe delivery of the actuator.

4.4.1.2 Level C. The actuator, packaged and packed in accordance with Section 5, Level C, or the contract or the purchase order, shall be vibrated in accordance with WR-11. Starting at 2 cps, gradually increase the frequency until the pack just begins to leave the table. At this point, vibrate for a period of one hour. Thereafter, the actuator shall meet the performance requirements of 3.3.

4.4.2 Vibration; ready nonoperating. The actuator, less pack and packaging, shall be mounted rigidly and secured in a horizontal position to a vibration device, utilizing the three mounting holes on actuator assembly (BUWEPS Drawing 694739). The actuator shall be vibrated vertically for a period of 3 hours at a frequency of 60 cps and a double amplitude (peak-to-peak) of 0.016 plus or minus 0.002 inch. At the conclusion of the test the actuator shall meet the performance requirements and product characteristics of this specification.

4.4.3 Shock, transportation.

4.4.3.1 Level A (multiple pack). The actuator, packaged and packed in accordance with Section 5, Level A, shall be dropped from a free fall height of 30 inches, in accordance with the requirements of WR-11, onto a nonyielding surface. Thereafter, the packed condition shall be examined for structural integrity to assure safe delivery of the actuator.

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4.4.3.2 Level C. The actuator, packaged and packed in accordance with Section 5, Level C, or the contract or the purchase order, shall be dropped from a free fall height of 30 inches, in accordance with the requirements of WR-11, onto a nonyielding surface. Thereafter, the actuator shall show no evidence of damage and shall meet all the performance requirements of 3.3.

4.4.4 Temperature and humidity

4.4.4.1 High temperature and humidity. The actuator, less pack and packaging, shall be placed in a test chamber and subjected to a temperature of plus 130 degrees plus or minus 5 degrees F, 90 to 95 percent relative humidity for a period of 48 hours. At the conclusion of this period the actuator shall be removed from the test chamber and allowed to return to and remain at room temperature for a period of four hours, after which the actuator shall meet the performance requirements and product characteristics of this specification.

4.4.4.2 Low temperature. The actuator, less pack and packaging, shall be placed in a test chamber and subjected to a temperature of minus 20 degrees F plus or minus 5 degrees F for a period of four hours. At the conclusion of this period the actuator shall be removed from the test chamber and allowed to return to and remain at room temperature for a period of four hours, after which the actuator shall meet the performance requirements and product characteristics of this specification.

4.4.5 Operating life

4.4.5.1 Run the actuator for 50 duty cycles, consisting alternately of runs as specified in 4.5.4.11.1, 4.5.4.11.2, 4.5.4.11.3, and 4.5.4.11.4, and cooling periods as specified in 3.3.10.4. At no time during the operating life tests shall the actuator torque exceed the limits specified in 3.3.8 or the oscillations exceed the limits of 3.3.9.2. Prior to the start of the test, the actuator test stand and spring load shall be calibrated to permit reading of the torque output during life testing.

4.4.5.2 At the conclusion of the 50th duty cycle and after cooling as specified in 3.3.10.4, make a run as specified in 4.5.4.7.

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4.4.5.3 The actuator shall be considered to have successfully withstood the test for operating life if, at the conclusion of this life test, it meets the requirements of 3.3 when tested as in 4.5.

4.5 Acceptance inspection. Each magnetic particle clutch actuator delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.2 Test conditions. Unless otherwise specified, the actuator shall be subjected to acceptance under the following conditions:

4.5.2.1 Temperature: 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.5.2.2 Altitude: Normal ground.

4.5.2.3 Vibration: None.

4.5.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.5.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.5.3.1 One stroboscopic tachometer capable of measuring speeds to at least 7000 rpm, with an accuracy of plus or minus 2 percent, and of operating from a 115 volt ac, 60 cps power source.

4.5.3.2 Recording oscillograph, one or more channels.

4.5.3.2.1 Chart speed of 5 or 10 and 100 or 125 millimeters (mm) per second.

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4.5.3.2.2 Chart width per channel of at least 40 mm.

4.5.3.2.3 Tolerance on chart speed plus or minus 0.5 percent or better, or the inclusion of a timing record to monitor chart speed.

4.5.3.2.4 Sensitivity. Ten millivolts per mm of deflection, or better.

4.5.3.2.5 Frequency response. Flat from dc to at least 15 cps.

4.5.3.3 One protractor capable of measuring angles up to at least 30 degrees, and of being read in increments of not greater than 0.2 degree.

4.5.3.4 One multimeter, AN/PSM-4, or equivalent.

4.5.3.5 One torque gage capable of measuring from zero to at least 60 pound-inch in increments of 1 pound-inch maximum, and with an accuracy of plus or minus 3 percent.

4.5.3.6 Test stand assembly, BUWEPS Drawing 877495, or equivalent, including springload assembled, BUWEPS Drawing 877168, and torque meter, BUWEPS Drawing 1155054.

4.5.3.7 Power sources.

4.5.3.7.1 115 volt ac, 60 cps, single phase.

4.5.3.7.2 One actuator adjustment set, BUWEPS Drawing 877336, including motor-driven potentiometer, to produce output that will simulate requirements for the duty cycle as specified in 3.3.10.

4.5.3.8 One stop clock capable of measuring time in increments of 1 second or less and recording a total time of at least 60 minutes.

4.5.3.9 One centigrade thermometer to include the range of 30 degrees C to plus 60 degrees C, calibrated in increments of 1 degree C with an accuracy of plus or minus 0.5 degree over the specified range.

4.5.3.10 One backlash measuring instrument capable of measuring angular movement up to at least 2 degrees in increments of not greater than 0 degrees, 1 minute.

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4.5.3.11 Megohmmeter capable of measuring insulation resistance at 500 volts dc with sufficient scales to permit readings in the middle third of the scale of 1 and 20 megohms, with a 2 percent accuracy on all scales.

4.5.4 Test procedure

4.5.4.1 Insulation resistance. Connect the megohmmeter (4.5.3.11) between ground and all pins of P40, connected together. Measure insulation resistance in accordance with NAVORD OD 12301. The resistance shall be as specified in 3.3.1.

4.5.4.2 Actuator circuit resistance.

4.5.4.2.1 Following assembly of the complete actuator, connect the positive lead of the multimeter (4.5.3.4) to terminal 2 and the negative lead to terminal 1 of connector P40 (BUWEPS Drawing 694739), and measure the aft coil resistance, which shall be as specified in 3.3.2.1.

4.5.4.2.2 To verify presence of the diode, parallel with aft coil, connect the positive lead of the multimeter to terminal 1 and the negative lead to terminal 2 of connector P40 (BUWEPS Drawing 694739). Set resistance selector switch to XI scale. The meter shall indicate on the resistance scale a value not greater than 20 or less than 10.

4.5.4.2.3 Following assembly of the complete actuator, connect the positive lead of the multimeter to terminal 2 and the negative lead to terminal 3 of connector P40 (BUWEPS Drawing 694739), and measure the forward coil resistance, which shall be as specified in 3.3.2.1.

4.5.4.2.4 To verify presence of the diode, parallel with forward coil, connect the positive lead of the multimeter to terminal 3 and the negative lead to terminal 2 of connector P40 (BUWEPS Drawing 694739). Set resistance selector switch to XI scale. The meter shall indicate on the resistance scale a value not greater than 20 or less than 10.

4.5.4.3 Binding

4.5.4.3.1 Input. With the actuator in a horizontal position (output arm down) and with the magnetic particle

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clutch coils deenergized, manually rotate the input shaft at least 20 turns in each direction. There shall be no evidence of tightness or binding, to meet the requirements of 3.3.3.1.

4.5.4.3.2 Output. With the actuator in a horizontal position (output arm down) and with the magnetic particle clutch coils deenergized, measure the breakaway torque of the output arm in both directions, using the torque gage (4.5.3.5). The breakaway torque shall be as specified in 3.3.3.2.

4.5.4.4 Output travel

4.5.4.4.1 Mount the actuator (output arm down) in the test stand (4.5.3.6) with the spring load disengaged.

4.5.4.4.2 Connect the actuator adjustment set (4.5.3.7.2) to connector P40 (BUWEPS Drawing 694739).

4.5.4.4.3 With the actuator input pulley running at 5000 to 6000 rpm in a cw direction (viewed facing the pulley), apply 12.5 plus or minus 0.5 ma in each clutch coil alternately, and measure the total travel between stops, using the protractor (4.5.3.3). The travel of the output shall be as specified in 3.3.4.

NOTE: Setting of input speeds. With the actuator mounted and driven as required for the condition being tested, focus the strobe light of the stroboscopic tachometer (4.5.3.1) on the input pulley of the actuator. Adjust the frequency of the strobe light until the pulley appears to be stationary. The pulley speed indicated on the stroboscopic tachometer dial shall correspond to the required input shaft speed.

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4.5.4.4.4 Input-output relationship. Mount the actuator as in 4.5.4.4.1 and connect it as in 4.5.4.4.2. Manually move the output arm against rear stop. Apply 7 plus or minus 2 ma to the forward coil, and deenergize the aft coil. Place adjacent marks on the input pulley and actuator housing. Manually rotate input pulley 19 turns. Using the protractor measure travel of output arm. Output arm shall move 24 to 26 degrees.

4.5.4.4.5 Repeat 4.5.4.4.4, with the exception that 7 plus or minus 2 ma is applied to aft coil and forward coil is deenergized. Output arm shall move 24 to 26 degrees.

4.5.4.5 Backlash

4.5.4.5.1 Output arm backlash

4.5.4.5.1.1 Mount the actuator as in 4.5.4.4.1.

4.5.4.5.1.2 Connect the actuator adjustment set as in 4.5.4.4.2.

4.5.4.5.1.3 Attach the backlash measuring device (4.5.3.10) to the output arm or the output shaft (BUWEPS Drawing 694739).

4.5.4.5.1.4 With the input pulley driven at 5000 to 6000 rpm, apply 5.75 plus or minus 0.75 ma to both coils.

4.5.4.5.1.5 Stop the input pulley rotation and lock it; measure the total angular freedom of backlash of the output shaft or arm. The backlash shall be as specified in 3.3.5.1.

NOTE: The input pulley shall not be permitted to rotate in reverse direction prior to locking.

4.5.4.5.2 Potentiometer shaft backlash

4.5.4.5.2.1 Rotate output shaft cw to approximate mid-position, and record resistance between pins 5 and 6 of P40 (BUWEPS Drawing 694739); also record angular position of output arm at this position. Continue cw rotation to stop.

4.5.4.5.2.2 Rotate output ccw until resistance between pins 5 and 6 is the same as the value recorded above.

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Record angular position of output arm. The maximum variation between the two recorded angular positions above (backlash) shall be as specified in 3.3.5.2.

4.5.4.6 Output direction

4.5.4.6.1 With the actuator mounted as in 4.5.4.4.1, the actuator adjustment set connected as in 4.5.4.4.2, and with the pulley driven at 5000 to 6000 rpm in a cw direction (as viewed facing the pulley), energize the aft coil (1 and 2 of P40) with 7 plus or minus 2 ma. The output arm shall move as specified in 3.3.6.1.

4.5.4.6.2 With the actuator mounted and driven as in 4.5.4.6.1, energize the forward coil (2 and 3 of P40) with 7 plus or minus 2 ma. The output arm shall move as specified in 3.3.6.2.

4.5.4.7 Output torque

4.5.4.7.1 Mount the actuator to the test stand (4.5.3.6) with the torque gage connected to the actuator output arm (the spring load is disconnected).

4.5.4.7.2 Connect the actuator adjustment set to connector P40 (BUWEPS Drawing 694739), and drive the input pulley at 5000 to 6000 rpm in a cw direction, as viewed from the pulley end.

4.5.4.7.3 Adjust the actuator adjustment set for “automatic” variation of control current so as to alternately apply 12.5 plus or minus 0.5 ma to each clutch coil. There shall be less than one ma in the other clutch coil. The measured torque output from the actuator in either direction shall be as specified in 3.3.8.1, to meet the requirement of 3.3.7.

4.5.4.8 Balanced current operation

4.5.4.8.1 With the actuator mounted and driven as specified in 4.5.4.7, adjust the actuator adjustment set so that when the currents of both coils are within 0.25 ma of being equal, the sum of the two currents shall be 11 plus or minus 2.5 ma. The torque shall not exceed the limits specified in 3.3.8.2, to meet the requirements of 3.3.7.

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4.5.4.8.2 Having obtained a balanced current condition, the ripple voltage measured across pins 13 and 14 of P40 shall not exceed 80 millivolts peak-to-peak.

4.5.4.9 Smoothness of operation

4.5.4.9.1 Unloaded output

4.5.4.9.1.1 Mount the actuator to the test stand with the output arm unloaded; connect the actuator to the actuator adjustment set and to the recording oscillograph (4.5.3.2) to record the position of the followup potentiometer, and drive the input pulley at 5000 to 6000 rpm. The input pulley speed shall be held constant within one-half percent during the period of actuator movement.

4.5.4.9.1.2 Alternately energize each clutch coil with a current of 7 plus or minus 2 ma, at the same time removing current from the other coil, and record the position of the potentiometer.

NOTE: The sensitivity of the oscillograph shall be adjusted for a band width of not less than 35 mm deflection between actuator stops, and the chart speed shall be adjusted so that the trace crosses the chart at an angle of 30 to 60 degrees during actuator travel.

4.5.4.9.1.3 Repeat 4.5.4.9.1.1 and 4.5.4.9.1.2 for the 625 ohm recording potentiometer.

4.5.4.9.1.4 The linearity of the actuator output, as determined from 4.5,4.9.1.2, shall be as specified in 3.3.9.1 to meet the requirements of 3.3.7.

4.5.4.9.1.5 Mount the actuator to the test stand with the output arm unloaded; connect the actuator to the actuator adjustment set and to the recording oscillograph to record tachometer output. Drive the input pulley at 5700 rpm. The input pulley speed shall be held constant within one-half percent during the period of actuator movement.

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4.5.4.9.1.6 Alternately energize each clutch coil with a current of 7 plus or minus 2 ma, at the same time removing current from the other coil. Record output of tachometer; it shall be in accordance with 3.3.4.1.

NOTE: The sensitivy of the oscillograph shall be adjusted for a band width of not less than 0.5 volt per millimeter and the chart speed shall be adjusted so that the maximum tachometer voltage trace is approximately 10 millimeters long. For convenience, this test may be performed for any constant input pulley speed between 5000 rpm and 6000 rpm, with the results corrected to 5700 rpm.

4.5.4.9.2 Loaded output at rated current

4.5.4.9.2.1 Mount the actuator to the test stand with the output arm attached to the spring load. Connect the actuator to the actuator adjustment set and to the recording oscillograph to record the position of the followup potentiometer, and drive the input pulley at 5000 to 6000 rpm.

4.5.4.9.2.2 Adjust the oscillograph sensitivity and chart speed as in 4.5.4.9.1.2; and adjust the actuator adjustment set to conform to the requirements of the duty cycle specified in 3.3.10.

4.5.4.9.2.3 Run through one complete duty cycle, recording the position of the output shaft, as indicated by the followup potentiometer. The extreme values of the irregularities of position, after the control current is at 12.5 plus or minus 0.5 ma for 1 second, shall be measured and converted into degrees. The total oscillation and irregularities during any one element of a duty cycle at rated current in each direction shall not exceed the limits specified in 3.3.9.2, to meet the requirements of 3.3.7.

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NOTE: It is not necessary to run recording paper through the oscillograph during the entire time of the performance cycle. It will be sufficient to record four complete elements at both the start and the end.

4.5.4.9.3 Loaded output at balanced current

4.5.4.9.3.1 Using the recording of the same performance cycle that was run for 4.5.4.9.2, the trace of the irregularities of position, after the control currents of the two magnetic particle clutch coils are equalized for one second, shall be measured and converted into degrees. The total oscillation during any one operation at balanced current shall not exceed the limits specified in 3.3.9.3, to meet the requirements of 3.3.7.

4.5.4.10 Performance with low input pulley speed

4.5.4.10.1 Adjust the pulleys and belts of the actuator and the test stand so that the actuator input pulley shaft will be driven at a speed between 1000 and 2000 rpm.

4.5.4.10.2 Check the input speed as described in the note following 4.5.4.4.3.

4.5.4.10.3 With the actuator input pulley shaft speed obtained in 4.5.4.10.1, repeat the performance tests of 4.5.4.7.1, 4.5.4.7.2 (except for input pulley speed), and 4.5.4.7.3.

4.5.4.10.4 The measured torque output from the magnetic particle clutch, in either direction, shall be as specified in 3.3.8.1, to meet the requirements of 3.3.7.

4.5.4.11 Duty cycle test

4.5.4.11.1 With the actuator mounted, connected, and running as specified in 4.5.4.9.2.1, throw the two DUMMY LOAD-CLUTCH switches on the actuator adjusting set to the DUMMY LOAD position, operating the ADJ POT MAN knob so that

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the current in each clutch is less than 3 ma when the switch for that clutch is operated. Switch the ADJ POT MAN-AUTO POT switch to AUTO-POT and, watching the meters, adjust the BAL CURRENT potentiometer and the two AUTO POT TRIM potentiometers so as to make the peak current in each direction fall within 12.5 plus or minus 0.5 ma. With these adjustments made, the currents shall be equal within 0.25 ma during the "balanced current" periods, as shown in Figure 2.

4.5.4.11.2 Switch the DUMMY LOAD-CLUTCH switches, one at a time, to CLUTCH, making the transition while the current in that side is less than 3 ma and noting the time,

4.5.4.11.3 Operate the actuator for a minimum of 30 minutes under this condition, and once during each 5 minute interval operate the oscillograph at the slowest chart speed for 1 or 2 cycles. There should be no excessive oscillation at rated or balanced current in either direction, nor should there be a complete loss of travel in either direction.

4.5.4.11.4 At the conclusion of 4.5.4.11.3, return each DUMMY LOAD-CLUTCH switch to the DUMMY LOAD position, in turn, while the current in that clutch is less than 3 ma, and return the ADJ POT MAN-AUTO POT switch to the ADJ POT MAN position.

4.5.4.11.5 Immediately after completion of a duty cycle, measure the insulation resistance as specified in 3.3.1, by the method indicated in 4.5.4.1.

4.5.4.11.6 Replace the spring load on the test stand with the torque measuring device, allow the actuator to cool within 5 degrees C (9 degrees F) of ambient, and repeat the tests specified in 4.5.4.7.

4.5.4.11.7 Compliance with the limits of the tests specified in 4.5.4.11,6 shall satisfy the requirements of 3.3,7.

4.5.4.12 Diode. Verify that the diode complies with requirements of 3. 3.2.3.

4.5.4.13 Tachometer. Verify that the tachometer complies with requirements of 3.3.4.2.

4.5.4.14 Packaging and packing. The actuator shall be examined to ensure that it complies fully with the requirements of Section 5.

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4.5.4.15 Waiver of tests. Tests specified in 4.4.1 (Vibration; transportation) and 4.4.3 (Shock; transportation) may be waived by the procuring agency for Level C packs only, provided the contractor guarantees the item will be delivered to the using activity in such a condition that any item selected shall be capable of meeting all the requirements of 3.3.

4.5.4.16 Acceptance criteria. Failure of the actuator, tachometer, or diode to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A

5.1.1.1 Cleaning. If further cleaning is required after final assembly cleanup, the actuator shall be cleaned in accordance with Method C-1 of Specification MIL-P-116, provided the process is not injurious to the item.

5.1.1.2 Drying. If required, the actuator shall be dried in accordance with one or more of the procedures given in Specification MIL-P-116, provided the process is not injurious to the item.

5.1.1.3 Preservative application. None.

5.1.1.4 Unit packaging. Each actuator shall be packaged in an individual metallic container in accordance with the drawings and other requirements listed in BUWEPS LD 495980 for actuator 694739-1 and LD 620137 for actuator 694739-2.

5.1.1.5 Intermediate package. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning, drying, and preservative application. Cleaning, drying, and preservative application shall be in accordance with 5.1.1.

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5.1.3.2 Unit packaging. Unit packaging shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one actuator.

5.1.3.3 Intermediate package. Not applicable.

5.2 Packing

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with Specification MIL-P-16298.

5.2.1.1 Exterior containers. Unless otherwise specified in the order or contract, the actuator, packaged as specified in 5.1.1.4, may be shipped individually or packed in an overseas type container in accordance with Specification MIL-P-16298 (gross weight not to exceed 200 pounds).

5.2.2 Level B. Not applicable.

5.2.3 Level C. The actuator, packaged in accordance with 5.1.3.2, shall be packed to afford protection against damage during direct shipment from the source of supply to the first receiving activity for immediate use. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation.

5.3 Level C, cushioning. The unit package and pack shall incorporate sufficient cushioning material, bracing, or other shock absorbing devices to ensure the item will meet all the requirements of Section 3 after being tested in accordance with 4.4.1.2 and 4.4.3.2.

5.4 Marking.

5.4,1 Special markings. None unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by contract, order or drawings, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including requirements for serial numbers on unit packages and shipping containers.

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6. NOTES

6.1 Intended use. The magnetic particle clutch actuator is intended for use in Torpedo Mk 37 Mods 0 and 1, or other suitable application.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 The specific title, number, and date of this specification and exceptions to this specification, applicable drawings and other documents.

6.2.2 The applicable levels of preservation, packaging, and packing.

6.2.3 That all additional samples required because of failure of any unit(s) to pass the prescribed periodic production examinations shall be provided and inspected at the expense of the supplier.

6.2.4 That certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Bureau of Naval Weapons when testing is accomplished at other than a Government activity.

6.3 Criteria for use of proper level of preservation, packaging, and packing:

6.3.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations, or stored under indeterminate conditions for redistribution anywhere.

6.3.2 Level B. This level shall be used only when it is definitely known that the item will be held in covered storage, either in domestic or overseas locations, for a period of 6 months or less.

6.3.3 Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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